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DSBA 6156 Final Project
UNC Charlotte

PREDICTING FLIGHT DELAY RISK FOR PASSENGERS TRAVELING THROUGH CHARLOTTE

BUSINESS CASE / EXECUTIVE SUMMARY

Predicting Flight Delay Risk for Passengers

Problem:

Flight delays can cause passengers to miss connections and disrupt travel plans, especially at major hubs like Charlotte.

Impact:

Missed connections lead to longer travel times, rebooking costs and poor passenger experience. Flight delays cost U.S. passengers an estimated \$28 billion annually in lost time, rebooking fees, and missed connections (FAA, 2023).

Approach:

Use machine learning models to predict whether an American Airlines flight departing from Charlotte Douglas International Airport (CLT) will be delayed.

End Users / Production Plan:

Passengers can use delay risk predictions when booking or planning trips to choose flights with lower delay risk or safer connection times.

Business Objective:

The goal of this project is to build a predictive model that estimates the probability that a flight will be delayed before departure.

DATA & MODELING APPROACH

Dataset:

U.S. Bureau of Transportation Statistics
Airline On-Time Performance Data.

Focus:

American Airlines flights departing from Charlotte Douglas
International Airport (CLT) using 2024–2025 data.

DATA & MODELING APPROACH CONTINUED

Key Features

- Scheduled departure time
- Day of week / month
- Destination airport
- Scheduled flight duration (minutes)

Target Variable

Delayed = 1 if departure delay > 15 minutes

Delayed = 0 otherwise

Modeling Technique

This project will use supervised classification models, including:

- Logistic Regression
- Random Forest
- Gradient Boosting

These models will be compared to identify which provides the best balance between predictive performance and interpretability

DEPLOYMENT & PROJECT TIMELINE

Deployment Plan

The model will generate predictions on the likelihood of flight delays.

Delay risk predictions will be displayed directly in flight search results within a booking platform (e.g., airline website or travel app). Predictions will run in batch using upcoming flight schedules.

Passengers could use this information to:

- Compare delay risk across flight options
- Avoid flights with high predicted delay probability
- Choose safer connection times

The model should be easy to interpret using feature importance. Since predictions will run in batch, latency isn't a concern, and there are no major regulatory requirements for this project.

Project Timeline

March 25 — Initial proposal presentation and incorporate feedback

April 1 — Data collection and exploration

April 10 — Feature engineering

April 15 — Model development and evaluation

April 20 — Documentation and final preparation

April 22 — Final project submission